

Hatim Mrabet

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Education

ENS Paris-Saclay - Master MVA

Master of Science - Mathematics, Vision and Learning

Paris, FR

2024–2025

Research-oriented master program.

Coursework: Generative Models for Images, Deep Learning, Stopping Times and Online Algorithms, Machine Learning for Time Series, Audio Signal Processing, Learning Applied to Text and Graph Data, Large Language Models

École CentraleSupélec - Engineering Degree

Master of Science - Mathematics and Data Science major

Paris, FR

2021–2025

Grade: 4.0

Coursework: Machine Learning, Computer Vision, RL, Advanced Probabilities, Optimization, Statistics, Partial Differential Equations, Algorithms and Data Structures, Object-Oriented Software Engineering.

Experience

Inria

Research Intern, **SCOOL** team

Lille, FR

Apr 2024–Aug 2024

- Fine-tuned a word embedding model using contrastive learning techniques inspired by **SimCLR**, improving food classification **hits@1** metric by **8%** (from **74%** to **82%**) through semantic search.
- Developing a Machine Learning model (**RandomForest**) to predict the Symptoms of IBS, given the diet followed by the patient.
- Developing a **bandit algorithm (UCB)** to propose the most adequate and efficient diet that must be followed by the patient to reduce the symptoms of IBS.

BNP Paribas

Data Science Intern

Paris, FR

Jul 2023–Jan 2024

- Developed machine learning pipeline for BII (Biodiversity Intactness Index) estimation. Tested regression models (**Linear regression, CatBoost, RandomForest**) for species abundance and regional similarity prediction.
- Constructed housing unit dataset and developed Machine Learning models (**Linear Regression, XGBoost, RandomForest**) for predicting energy performance diagnosis (DPE) based on unit features.

Projects

Research project : Image segmentation of the Optic Nerve Sheath (few shots learning problem)

- Employed Deep Learning techniques (**ResNet and Unet**) for image segmentation and feature extraction.
- Utilized clustering methods (**HDBSCAN and Kmeans++**) for image classification.
- Used **Transfer learning** by training the decoder of a pretrained Unet on our dataset

Creative Image Generation with GANs

- Developed advanced **Generative Adversarial Networks (GANs)** using diverse architectures, including **CNN-based models, U-Net-based discriminators**, and **U-Net decoder-based generators**.
- Integrated **CutMix augmentation** and **consistency regularization** techniques to boost feature learning and model generalization.
- Conducted thorough evaluation and comparison of model performance across **multiple datasets** (Paintings, Pixel Art).

Movies Recommendation System using Bandit Algorithms

- Cleaned and preprocessed a movie ratings dataset, extracting features to enrich the data.
- Developed and compared **contextual bandit** algorithms based on **UCB** and **G-optimal design for exploration**, evaluating performance with varying feature sets and model arms.
- Implemented and assessed **three multi-armed bandit models (UCB, Thompson Sampling, EXP3)** across different setups by varying the number of arms.

Skills

Technical Skills: Python(Tensorflow,PyTorch,Keras,Scikit-learn, ChromaDB, Sentence-Transformers...),SQL,Java,Matlab.

Tools: VSCode, Eclipse, Git, Jupyter,Latex, Domino.

Languages: English (Fluent) | French (Fluent) | Arabic (Native)